



Predicting soil water movement in converted soybean fields under high moisture condition

Chihiro Kato¹ · Choichi Sasaki¹ · Akira Endo¹ · Nobuhiko Matsuyama¹ · Taku Nishimura²

Received: 29 October 2018 / Revised: 13 December 2018 / Accepted: 27 December 2018 / Published online: 29 March 2019
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Abstract

Crops adapt to variable soil moisture by varying the distribution of their roots and patterns of water absorption. Meanwhile, proper numerical model for predicting soil moisture in croplands especially under high soil moisture conditions is still under discussion. In this study, we evaluated a numerical model for predicting soil moisture in converted soybean fields. A soybean-growing experiment with a converted field model was conducted under the conditions of (a) “G40,” in which the groundwater level was at a depth of 40 cm from the soil surface during the whole experimental period, and (b) “G10-40,” in which there was overmoisture during the early growing stage. Then, a numerical simulation of soil water movement for this growing experiment was conducted with the HYDRUS-1D model. At the end of this experiment, approximately 75% and 60% of the total root mass was concentrated in the topsoil layer (0–10 cm depth) of G10-40 and G40, respectively; thus, the distribution of soybean root might be strongly affected by soil moisture at the early growing stage. The measured soil water consumption rate was relatively high around 30–90 days after seeding in G40, while around 60–90 days after seeding in G10-40. The simulation results improved with the consideration of root distribution variation, changes in evapotranspiration rate and LAI values. The dual-porosity model might be appropriate for describing the soil hydraulic characteristics of soils with macropores, but further improvement is needed in determination of the parameters for high soil moisture.

Keywords Soil moisture · Converted fields · Numerical simulation of soil water movement · HYDRUS-1D · Excess water

Introduction

Low permeable clayey soil and/or shallow groundwater table in fields converted from paddies often induce water-logging that results in poor yield. Furthermore, recently in Japan, increases in storm frequency and single-event rainfall intensity have been observed (Japan Meteorological Agency 2018), which mean that the risk of excess water in fields has been growing. Therefore, simulating and controlling field soil moisture are required to assure a stable yield and the proper quality of products.

Production of soybean, the major crop of converted fields in Japan, is susceptible to excess soil water, which typically

occurs during the early stage of vegetative growth in the rainy season of Japan. Crops adapt to variable soil moisture, such as during periods of excess water and drought, by varying the distribution of their roots and patterns of water absorption (e.g., depth, rate, timing etc.) (Hirasawa et al. 1994). Meanwhile, a proper numerical model for predicting soil moisture in croplands especially when under nearly saturated conditions is still unavailable and under discussion. In this study, we evaluated a numerical model for predicting soil moisture in clayey converted soybean fields.

Materials and methods

Growing experiment

A soybean-growing experiment with a converted field model was conducted in 2017. Two plastic containers (W:52.5 cm, D:34 cm and H:56.4 cm) in a glasshouse of Hirosaki University, Aomori, Japan, were used to monitor soil moisture with growing soybean. A clay loam

✉ Chihiro Kato
chihirok@hirosaki-u.ac.jp

¹ Faculty of Agriculture and Life Science, Hirosaki University, 3, Bunkyo-cho, Hirosaki, Aomori 036-8561, Japan

² Graduate School of Agricultural and Life Sciences, The University of Tokyo, 1-1-1, Yayoi, Bunkyo-ku, Tokyo 113-8657, Japan

soil sample for this growing experiment was collected from Kanagi farm of Hirosaki University, Goshogawara, Aomori, Japan. (Hereafter, we call this soil “Kanagi soil.”) Before packing, air-dried soil aggregates with 20–30 mm in diameter were artificially made with Kanagi soil. Then, from the bottom of each container, gravel ($\phi = 2\text{--}4.8$ mm; dry bulk density $\rho_d = 1.5 \text{ g cm}^{-3}$; 10 cm thickness) and aggregated Kanagi soil ($\rho_d = 0.89 \text{ g cm}^{-3}$, upper 40 cm thickness) were packed. Before seeding, irrigation was conducted to attain moisture of Kanagi soil to be proper for germination. “Ryuhō” cultivar, widely grown in northeast (Tohoku) region of Japan, was selected for this experiment. Six soybean seeds were planted at the center of each container on June 3rd and four plants out of the six were thinned out on June 19th; in other words, the two well-growing soybean plants per container were left for the experiment. During the period from the seeding to the thinning, the groundwater level of both containers was controlled at 40 cm in depth from the soil surface, in order to ensure germination and rooting of the plants. After thinning out, the soybean plants which were left in the containers were grown under the conditions of (a) optimum (in which the groundwater level [GWL] was at a 40 cm depth from the soil surface during the whole experimental period; hereinafter, “G40”), and (b) excess moisture during the early growing stage (in which GWL was regulated at 10 cm in depth from the soil surface for the first 40 days from the thinning and at 40 cm in depth from the soil surface for the following period; hereinafter, “G10-40”). This period, from the middle June to late July, corresponds to the typical rainy season in northern Tohoku region (on average, from June 14th to July 28th). Groundwater level was regulated by installing a Marriott supplying tank and drainage tank. The soybean plants were harvested on September 27th for G10-40 and October 5th for G40. Before thinning out, the soil surfaces were moistened accordingly, and after that, water was irrigated in accordance with the average rainfall pattern of Hirosaki City; specifically, 7.2 mm every 5 days, 13.1 mm every 4 days, 13.4 mm every 3 days, and 13.9 mm every 3 days in June, July, August and September, respectively. The water was supplied around 10:00 and 16:00 of each day.

After harvesting, soil was sampled from each soil layer and then basic physical properties were measured (Table 1).

To monitor soil moisture continuously, soil moisture sensors (EC5, METER Inc., USA) were inserted at depths of 3, 7, 15, 25 and 37.5 cm from the soil surface and the data were recorded every 20 min. Since artificial aggregates were comparatively large, each soil moisture sensor was inserted into a lump of artificial aggregates to avoid blank part of the soil along with the sensor. The sensors were calibrated with Kanagi soil in the same way as Kato and Nishimura (2017). The groundwater level was measured with two manometers per container twice a day. Every day, the amount of supplied water (both irrigation and flux from the Marriott tank) and drained water was measured. The balance between these supplied flux and drained flux denotes the amount of evapotranspiration and stored water, which implies water consumption rate.

Root mass distribution

The stems of the soybean plants were cut off 2 cm above the soil surface for harvesting, and then soil blocks of $W:32 \text{ cm} \times D:34 \text{ cm} \times H:10 \text{ cm}$ (depths of 0–10 cm, 10–20 cm, 20–30 cm and 30–40 cm from the surface) around two soybean plants were grubbed up. Soil was immediately washed off in order to collect the roots from each block. Then the roots were oven-dried under 70°C for 1 week and weighed.

Numerical simulation

The HYDRUS-1D version 4.17 (Šimunek et al. 2013) model was used for predicting soil moisture. The governing equation of soil water movement was expanded using Richards' equation with root water uptake [Eq. (1)],

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left[K \left(\frac{\partial h}{\partial z} + 1 \right) \right] - S(h) \quad (1)$$

$$S(h) = \gamma(h) S_p \cdots S_p = b(z) T_0$$

where θ is the total volumetric water content [$\text{L}^3 \text{L}^{-3}$], t is time [T], z is downward positive spatial coordinate [L], h is pressure head [L], T is temperature [K], K [L T^{-1}] is the hydraulic conductivity for liquid-phase fluxes due to gradients in h , and $S(h)$ is root water uptake, which is described

Table 1 Comparison of basic soil physical properties between G40 and G10-40 after harvesting

Depth (cm)	G40			G10-40		
	K_s (cm s^{-1})	Porosity (%)	Dry bulk density (g cm^{-3})	K_s (cm s^{-1})	Porosity (%)	Dry bulk density (g cm^{-3})
5	2.11×10^{-2}	64.3	0.98	2.24×10^{-2}	62.8	1.03
15	3.51×10^{-3}	58.6	1.07	5.09×10^{-3}	61.3	1.05
25	1.90×10^{-3}	62.3	1.00	1.22×10^{-3}	60.4	1.07
35	1.19×10^{-3}	56.3	1.06	6.80×10^{-4}	56.4	1.19

as multiplication of the root water uptake stress response function $\gamma(h)$ and potential root water uptake rate S_p . Here, Feddes model (Feddes et al. 1974) was employed as $\gamma(h)$. We used the same parameters for $\gamma(h)$ as shown in Kato and Nishimura (2017). Potential root water uptake rate S_p can be described with the normalized water uptake distribution $b(z)$ and potential transpiration rate T_0 . Here, $b(z)$ was determined by the results of root mass distribution.

The calculation area was set to be 0–60 cm in depth from the soil surface and consisted of a 40-cm-thick aggregated Kanagi soil layer and 20-cm-thick gravel layer. Furthermore, the Kanagi soil layer was divided into two layers (from the surface to 10 cm was called “Kanagi-1” and the rest was called “Kanagi-2”) by considering the difference of final dry bulk densities caused by ramming for packing (Table 1). In determination of soil hydraulic properties, the dual-porosity Durner–Mualem model [Eq. (2)] was employed for Kanagi soil. Here, a digital agricultural land soil map provided by the Institute for Agro-Environmental Sciences, Japan (Takata et al. 2009, 2013), was used for identifying the soil type of Kanagi soil. Then, averaged water retention data of Kanagi soil were obtained from the “Agricultural soil profile physical properties database, Japan (SolphyJ; Eguchi et al. 2011).”

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r} = w_1 (1 + |\alpha_1 h|^{n_1})^{-m_1} + w_2 (1 + |\alpha_2 h|^{n_2})^{-m_2}$$

$$K(S_e) = \frac{K_s (w_1 S_{e_1} + w_2 S_{e_2})^l \left(w_1 \alpha_1 \left[1 - \left(1 - S_{e_1}^{m_1} \right)^{m_1} \right] + w_2 \alpha_2 \left[1 - \left(1 - S_{e_2}^{m_2} \right)^{m_2} \right] \right)^2}{w_1 \alpha_1 + w_2 \alpha_2} \quad (2)$$

where θ_r is the residual volumetric water content [$L^3 L^{-3}$], θ_s is the saturated volumetric water content [$L^3 L^{-3}$], K_s is the saturated hydraulic conductivity [$L T^{-1}$], S_e is the effective saturation [dimensionless], α_i , l , m_i ($= 1 - 1/n_i$) and n_i are empirical parameters [dimensionless] and w_i is weighing factors.

First, the water retention data from SolphyJ were fitted with the van Genuchten model by using the RETC program (Yates et al. 1992) and obtained θ_s , θ_r , α and n (Kato and Nishimura 2017), and those parameters are employed for θ_s , θ_r , α_2 and n_2 , respectively, in Eq. (2). Saturated hydraulic conductivity K_s was estimated by using the neural network Rosetta (Schaap et al. 1998) and l was assumed to be 0.5 (Kato and Nishimura 2017). In order to reflect the permeability of macropores, α_1 and n_1 were determined after Akay et al. (2008). The weighing parameters w_1 and w_2 were determined by trial and error, and 0.3 and 0.5 were employed for “Kanagi-1” and “Kanagi-2,” respectively. For hydraulic properties of the gravel layer, van Genuchten–Mualem model [in the case that w_2 in Eq. (2) is zero] was employed

and the parameters were determined with “Sand” of a soil catalog (Carsel and Parrish 1988).

Initial and boundary conditions for the simulation

Numerical simulation was conducted to calculate soil water movement for 120 days after seeding during the growing experiment. For the initial condition, GWL was set to be 40 cm from the surface, and we assumed that soil moisture above the groundwater was in equilibrium. The boundary condition (BC) of the bottom was pressure head in accordance with the set GWL. The BC of the top was given as precipitation rate, or irrigation intensity and evapotranspiration rate. Here, irrigation rate was converted as one-hour precipitation started from 10:00 and 16:00. In this study, potential evapotranspiration rate ET_p was obtained by multiplication of FAO reference evapotranspiration rate and crop coefficient K_{crop} . For K_{crop} , the Ministry of Agriculture, Forestry and Fisheries of Japan (MAFF 1997) proposed $K_{crop} = 0.7$ in the seeding and planting period, 1.1 for the vegetative period and 0.5 for the full-ripe and harvest period. Here, we determined the period of each growing stage with the changes in water consumption rate. In order to separate ET_p into potential evaporation rate E_p and potential transpiration rate T_p , the leaf area index (LAI) value was used for the calculation (Šimuněk et al. 2013). LAI was referred to in Bajgain et al. (2015) and Furuhashi et al. (2011), and its

periods for each growing stage were determined considering the growth situation.

Results and discussion

Figure 1 shows the measured GWL during the experiment. The average GWL of G40 was 39.6 cm from the surface through growing period (Fig. 1a) and that of G10-40 was 11.1 cm from June 20th to July 26th and 39.9 cm after July 26th (Fig. 1b). Thus, the GWL was well maintained with the planned value.

The total root mass—in other words, the sum of the root weights of two soybean plants which were distributed from the surface to 40 cm in depth in the sampling area (W 32 cm × D 34 cm)—was 28.02 g and 15.49 g in G40 and G10-40, respectively. Figure 2 shows a comparison of the root distribution between G40 and G10-40. In G10-40, approximately 75% of the total root existed in the top 10 cm, while in G40, 60% was distributed in the top layer and about

Fig. 1 Measured groundwater levels during the experiment of **a** G40 and **b** G10-40

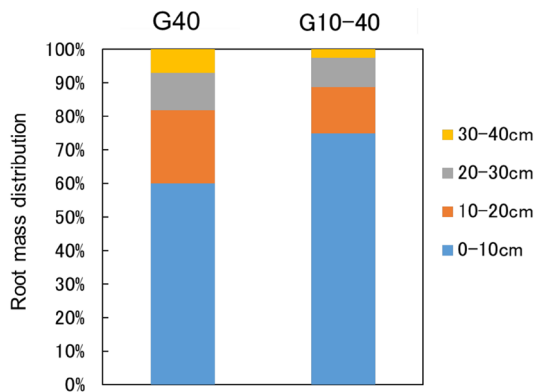
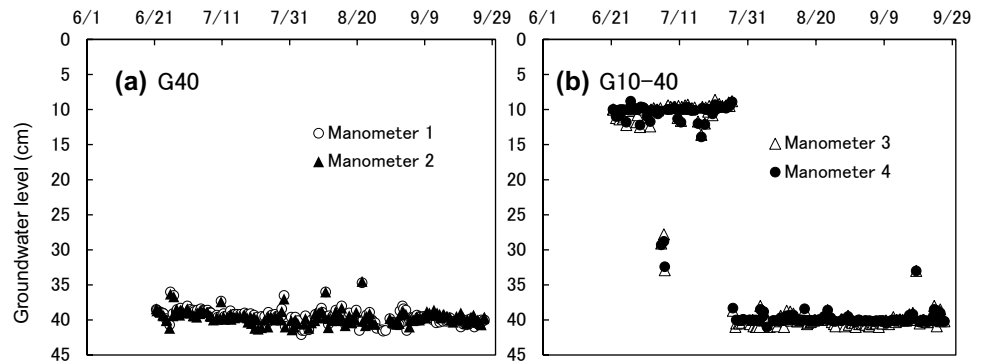


Fig. 2 Comparison of the root mass distribution between G40 and G10-40

10% of the roots penetrated to the depth of 40 cm. Haque et al. (2014) mentioned that 90% of total roots were observed in the top 10 cm layer when they were grown under the condition of $\text{GWL} = 10$ cm depth from the soil surface during the whole growing period. This implies that the root distribution could be affected by soil moisture in the early growing period.

Based on this root mass distribution, we determined the root water uptake rate distribution, $b(z)$ in Eq. (1), for the numerical simulation of the first 90 days as shown in Fig. 3a. Although, in this experiment, the soybean roots were sampled from 0 to 40 cm in depth, some fine roots were observed in the deeper layer and thus, in the numerical simulation, we assumed that the roots had penetrated to the depth of 50 cm as shown in Fig. 3a. Coelho and Or (1999) measured corn root water uptake (RWU) and root length density (RLD) patterns under surface and subsurface drip irrigation at different growth stages. They reported that RWU and RLD were almost similar, but RLD variation appeared to time lag behind the changes in RWU patterns and this difference might be caused by the very fine new roots, which could not be counted for RLD measurement. Here, since the soil moisture of the top layer increased in harvesting season as stated below (Fig. 7), it seemed to be caused by the decrease

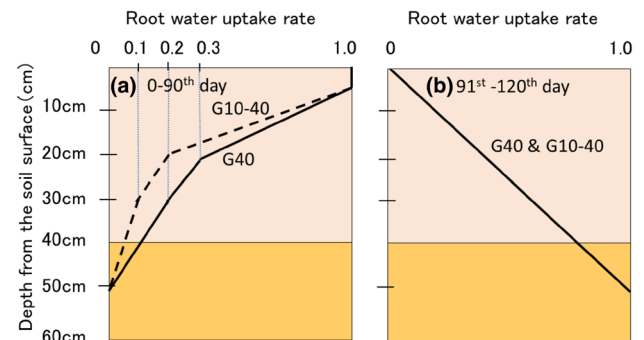


Fig. 3 Determined root water uptake rate distribution for numerical simulation **a** from 0 to 90th day and **b** from 91st to 120th day

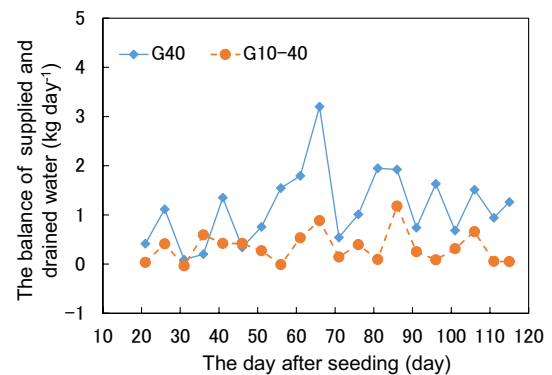


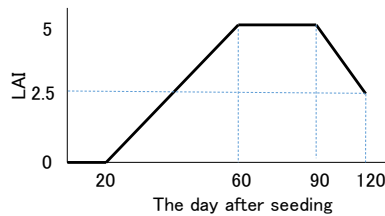
Fig. 4 Comparison of changes in the balance between supplied and drained water of the G40 and G10-40

in root water uptake especially in the surface layer. The distribution of root water uptake rate is determined in Fig. 3b for the simulation from the 91st to 120th day.

Figure 4 shows the changes in the balance between supplied and drained water of G40 and G10-40. In this figure, each data means five days' average. These values qualitatively imply the soil water consumption rate. The values of G40 were larger than those of G10-40. This might also be expected from the result that the plant height of G40 was

Table 2 Crop coefficient used for the simulation

Growing stage	Crop coeff. (MAFF 1997)	The day after seeding	
		G40	G10-40
Seeding and planting period	0.7	0–30th	0–60th
Vegetative period	1.1	31st–90th	61st–90th
Full-ripe and harvest period	0.5	91st–120th	91st–120th

**Fig. 5** Changes in LAI values used for the simulation

relatively higher than that of G10-40 through the experiment period. For example, the average plant height of G40 and G10-40 was 78.8 cm and 71.0 cm, respectively, on the 45th day (flowering period), and 114.9 cm and 98.8 cm, respectively, on the 65th day (data not shown). In G40, the values were relatively greater from the 30th to the 90th day, while in G10-40, the greater values appeared from the 60th to 90th day. From this result, crop coefficients and LAI values for the simulation are determined as shown in Table 2 and Fig. 5, respectively.

Determined soil hydraulic properties are shown in Fig. 6a water retention $\theta(h)$ and (b) hydraulic conductivity $K(\theta)$. Figure 7 shows the observed and simulated volumetric water content at depths of 3 and 25 cm from the soil surface in (a) G40 and (b) G10-40. In G10-40, observed soil moisture at a depth of 25 cm for the first one month from the seeding was missing because of the sensor trouble. During the period that the groundwater level was regulated at a depth

of 40 cm from the soil surface, the soil at a depth of 25 cm from the soil surface was unsaturated. Since Kanagi soil was clayey, it could happen that hydraulic conductivity of the lump of aggregates around the sensor as well as other artificial aggregates was low, and thus unsaturated though all the coarse inter-aggregate pores were saturated under groundwater table. During the same period, soil moisture at a depth of 10 cm rose with the increase in irrigation intensity and frequency. Observed soil moisture at depths of both 3 and 25 cm from the soil surface in G40 and at a depth of 3 cm in G10-40 increased after middle of September, or around the 90th day after seeding. This increase agreed with the leaf color change (from green to yellow; data not shown) and thus the evapotranspiration rate could decrease in this period. Regarding G40, the numerical model could describe the increase with rainfall and following decrease in soil moisture well, especially from June to the early September. Figure 8 shows changes in soil moisture at a depth of 3 cm from the soil surface that were simulated without considering crop coefficient and LAI (Kato and Nishimura 2017). Comparing Figs. 7 and 8, consideration of changes in crop coefficient and/or LAI at each growth stage improves the simulation results especially in the early growth and drying period in August.

Although, in this simulation, we assumed that the root water uptake rate at the soil surface layer and the evapotranspiration rate decreased in the harvest period (from the 90th to 120th day after seeding; Fig. 3, Table 2), the numerical model did not describe the increase in soil moisture of this period in both G10 and G10-40. In addition, the model underestimated and overestimated the soil moisture at depths of 3 and 25 cm, respectively, in July when GWL was set to be 10 cm from the soil surface. As well, the simulated results showed excessively rapid soil water drainage compared to the observed phenomena. These disagreements might be due to less accuracy in the estimation of soil hydraulic function at high soil moisture, which is one of the weak points of the Durner–Mualem model. In this study, artificially aggregated clayey soil was used for the

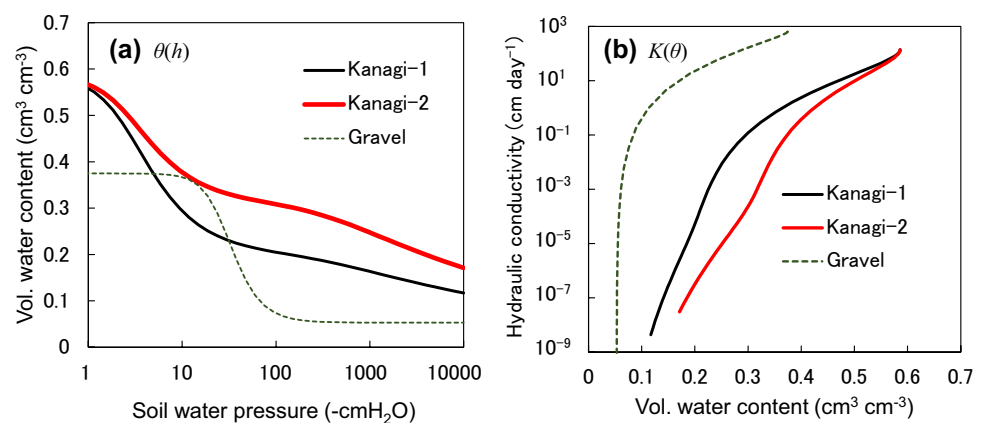
Fig. 6 Determined soil hydraulic properties for the simulation **a** $\theta(h)$ and **b** $K(\theta)$ 

Fig. 7 Comparison of simulated and observed changes in soil moisture at depths of 3 cm and 25 cm in **a** G40 and **b** G10–40

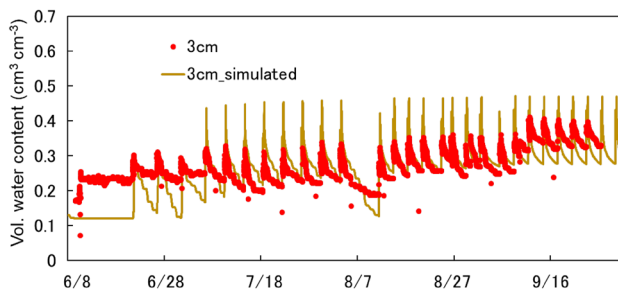
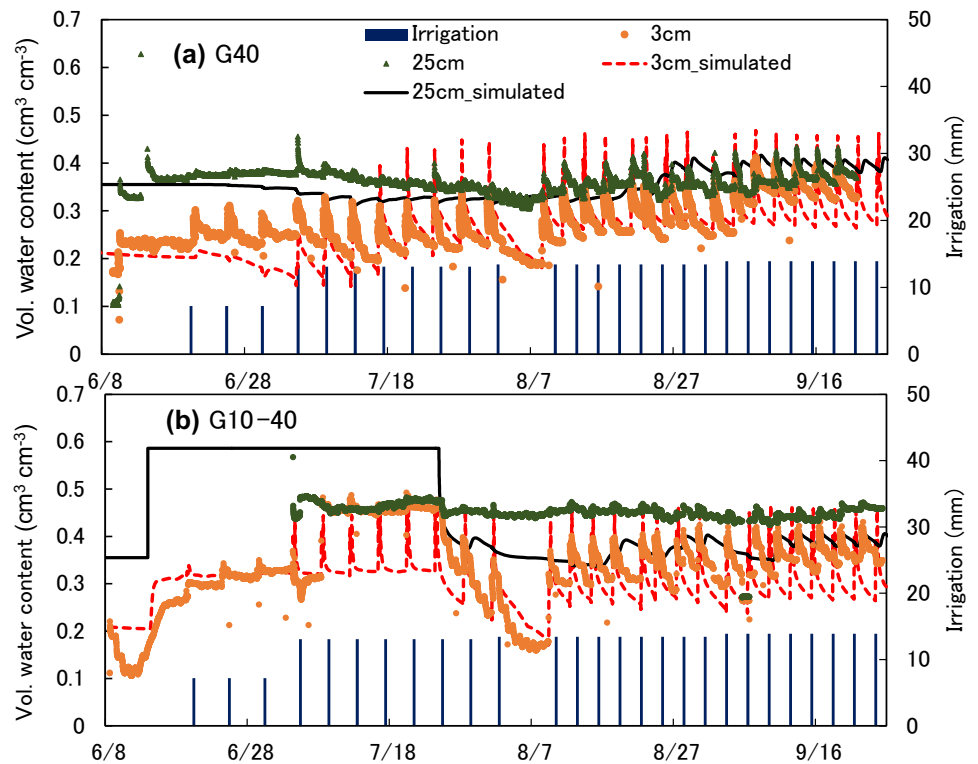


Fig. 8 Changes in soil moisture at depths of 3 cm in G40 simulated without consideration of crop coefficient and LAI

experiment in order to reproduce the field condition of typical converted upland fields. The Durner–Mualem model was useful for simulating soil water movement that was strongly affected by macropores and soil structure, and thus, the fluctuation of soil moisture was effectively described in this study when the soil moisture was relatively low. However, further discussion of soil hydraulic parameters should be conducted to improve the simulation's results.

Conclusion

In this study, we evaluated a numerical model for predicting soil moisture in clayey converted soybean fields with a growing experiment and numerical simulation. The final

root mass distribution of soybean was strongly affected by soil moisture condition in early growth. Roots tended to concentrate in the surface soil layer when soil moisture was relatively high. The soil water consumption rate changed with the soybean-growing stage. The maximum soil water consumption rate tended to appear late when the soybean plants were grown under high moisture conditions in early growth. The HYDRUS 1D model was effective in describing the changes in the soil water content of converted soybean field models by considering soybean root distribution and changes in evapotranspiration rate and LAI. To improve the simulation results, further modification of soil hydraulic function, especially at high water content, is needed.

Acknowledgements This study was conducted with the support of JSPS KAKENHI (Grant No. JP15K18756). The authors wish to thank the Tohoku Agricultural Research Center of Japan for providing soybean seeds. As well, we would like to thank Mr. Rikuto Toda, a graduate of the Faculty of Agriculture and Life Science, Hirosaki University, and Dr. Songtao Li, a graduate of the United Graduated School of Agriculture, Iwate University, for their help in conducting growing experiments.

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